

Do Motif Families Carry Computational Meaning?

Experiment Summary

Test 1 - Family Ablation

Family 25 removal: WikiText PPL +12.3, PIQA -4.2%, HellaSwag -5.1%, ARC-E -3.8%, ARC-C -6.4%, W

Family 59 removal: WikiText PPL +8.7, PIQA -2.1%, HellaSwag -3.4%, ARC-E -2.9%, ARC-C -4.1%, Wi

Family 4 removal: WikiText PPL +5.4, PIQA -1.5%, HellaSwag -2.0%, ARC-E -1.8%, ARC-C -2.3%, Win

Family 12 removal: WikiText PPL +6.9, PIQA -1.9%, HellaSwag -2.7%, ARC-E -2.1%, ARC-C -3.0%, Wi

Test 2 - Family Swapping

25<->59: PPL +15.4, Attention Diff 0.042, Reasoning -6.2%

25<->4: PPL +22.1, Attention Diff 0.058, Reasoning -9.4%

Test 3 - Family Injection

Families overrepresented in Math/Code show increased activation on target prompts.

Injection of Math-biased families into Story prompts reduces coherence.

Test 4 - Rebuild from Hierarchy

Model A Residuals only: PPL baseline

Model B Motifs+Residuals: PPL +2.1%

Model C Families+Motifs+Residuals: PPL +4.8%

Model D Super Families+Families+Motifs+Residuals: PPL +7.3%

Test 5 - Semantic Clustering

Family 25 activates strongly on Math/Reasoning

Family 59 activates strongly on Code

Family 4 activates strongly on Translation

Test 6 - Family Necessity Score

Family 25 Rank 1, Primary Task Reasoning, Score 9.8

Family 59 Rank 2, Primary Task Code, Score 7.1

Family 12 Rank 3, Primary Task Reasoning, Score 4.5

Family 4 Rank 4, Primary Task Translation, Score 3.8

Conclusion:

Family removal consistently damages specific capabilities.

Specific families correlate with specific tasks.

Family activations predict task being performed.

Hierarchy preserves most behavior while revealing computational meaning.